Daeyoung Roh

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Research Interests

Decision-aware LLM systems — reliability without wasted computation: *when to retrieve, read, verify, stop, or use a smaller model.* I study how to make LLM-based systems trustworthy at runtime *without retraining the model*: (i) **runtime control** of retrieval-augmented agents — deciding *when an agent should stop searching* and *whether it inspected evidence before answering*; (ii) **trajectory-level evaluation** beyond final-answer accuracy, decomposing where and why agents fail; (iii) **efficient system designs** that confine expensive LLM reasoning to offline stages while keeping the serving path lightweight, controlled, and verifiable.

Education

KAIST (Korea Advanced Institute of Science and Technology) Daejeon, Korea
M.S. in Data Science · GPA 3.65/4.3 (93.5/100) · Advisor: Prof. Mun Yong Yi *Sep. 2022 – Aug. 2024*

Handong Global University Pohang, Korea
B.A. in Economics & B.S. in Information and Communication Technology *Mar. 2018 – Aug. 2022*
GPA 3.59/4.5 (Major 3.88/4.5)

Publications

Under Review * denotes equal contribution

Titles of papers under double-blind review are withheld to preserve anonymity. The full CV, with titles and summaries, is available upon request.

1. **Two first-author papers** on the runtime reliability and efficiency of retrieval-augmented search agents.
   Daeyoung Roh, Donghee Han. *Under review at EMNLP 2026.*
   When a search agent should stop, and whether it inspected evidence before answering — runtime control of frozen agents, evaluated at the trajectory level.
2. **Two co-first-author papers** on efficient LLM-driven recommendation.
   Donghee Han*, **Daeyoung Roh***, et al. *Under review at CIKM 2026.*
   Confining LLM reasoning and profiling to offline stages so that query-time serving remains LLM-free and lightweight.

Peer-Reviewed

3. **Distilling LLM Reasoning into Dense Encoders: Bridging the Accuracy-Efficiency Gap in Recommendation**.
   Donghee Han, **Daeyoung Roh**, A Young Kim, Hwanjun Song, Mun Yong Yi. *Findings of ACL 2026.*
   Distills LLM reasoning into a 0.3B dense encoder via oracle-guided supervision; the LLM-free student outperforms its 12B teacher while cutting inference latency by an order of magnitude.
4. **Fine-Grained Multi-Prompt Essay Scoring with Multi-Level Disentanglement**.
   Donghee Han, **Daeyoung Roh**, Euihwan Han, Hwanjun Song, Mun Yong Yi.
   Data Mining and Knowledge Discovery (ECML-PKDD Journal Track), 2025.
5. **CTIP: Towards Accurate Tabular-to-Image Generation for Tire Footprint Generation**.
   Daeyoung Roh, Donghee Han, Jihyun Nam, Youngbin You, Jeongheon Park, Jungsoo Oh, Mun Yong Yi.
   IEEE/CVF WACV 2025.
6. **Closer through Commonality: Enhancing Hypergraph Contrastive Learning with Shared Groups**.
   Daeyoung Roh, Donghee Han, Daehee Kim, Keejun Han, Mun Yong Yi. *IEEE International Conference on Big Data 2025.*
7. **SIG-Net: GNN-based Dropout Prediction in MOOCs Using Student Interaction Graph**.
   Daeyoung Roh, Donghee Han, Daehee Kim, Keejun Han, Mun Yong Yi. *ACM/SIGAPP SAC 2024.*

Industry Experience

Ensapia (formerly Cocone), AL Lab Seoul, Korea
AI Engineer *Jul. 2024 – Present*

- Designed and operate a production **multi-agent RAG system** for enterprise knowledge search over heterogeneous sources (documents, internal portals, Google Workspace), serving diverse questions across departments; built a **feedback-driven knowledge-update pipeline**. Observations of hallucination and multi-hop failures in this system directly motivated my first-author research on agent runtime control.

- Built an **HR AI analysis platform** (STT+LLM interview evaluation) deployed in production for Korean and Japanese HR teams, and a **legal AI compliance tool** combining image-similarity search with LLM reasoning, used by the legal team.
- Led the **design-generation platform**: fine-tuning, inference serving, and operations for Flux-based image generation on AWS EKS / RunPod.

Research Experience

KAIST Knowledge Innovation Research Center (KIRC) Daejeon, Korea
Project Manager, Industry Collaboration with Hankook Tire *Nov. 2023 – Jun. 2024*

- Led tabular-to-image generation for tire footprint prediction; developed CTIP (tabular representation learning + Stable Diffusion), published at WACV 2025.

KAIST Knowledge Innovation Research Center (KIRC) Daejeon, Korea
Project Manager, Veterinary Diagnosis Prediction *Jul. 2023 – Dec. 2023*

- Built tree-ensemble models predicting ∼20 canine diagnoses from blood tests under small, imbalanced data.

Fount Inc., AI Core Department Seoul, Korea
Research Intern — ML for stock-price prediction *Jul. 2021 – Aug. 2021*

Handong Global University Pohang, Korea
Undergraduate Research Assistant *2021 – 2022*

- Financial-product risk analysis and monitoring system (Financial Supervisory Service project); ML-based regional housing-market analysis and fraud detection (Korea Real Estate Board project).

Honors & Awards

- **Excellence Award**, Hankook & Company Group–KAIST AI Competition, 2023 — tire-footprint generation; developed into the WACV 2025 paper.
- **Creativity Award**, K-DS Conference Hackathon, 2023 — RAG-based Korean-history chatbot with a thesaurus-grounded vector DB.

Teaching & Leadership

- Teaching Assistant for four courses, Handong Global University (2021–2022): Macroeconomics, Econometrics, Intro. to Economics, Intro. to Data Application.
- Editor-in-Chief (2019; Staff Reporter, 2018), *Handong Shinmun* (student newspaper) — led ∼20 student reporters: writing training, budget management, print/video editorial planning.

Technical Skills

- **ML / LLM systems:** PyTorch, HuggingFace (transformers/peft/trl), vLLM, LangChain, LoRA fine-tuning, retrieval (BM25/dense/FAISS), agentic RAG.
- **Statistical analysis:** paired bootstrap & non-inferiority testing, Holm correction, McNemar / permutation tests (used throughout my agent-evaluation work).
- **Infrastructure:** AWS EKS, RunPod, Docker, PostgreSQL, FastAPI, GPU serving pipelines.